

Aswin R Nair

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Education

Integrated Masters in Science (BS-MS), Physics

2020 – 2025

Indian Institute of Science Education and Research (IISER), Mohali, India

CPI 8.0 / 10

Relevant coursework: Machine Learning and Algorithms, High-Performance Computing (MPI), Numerical Methods in Computational Physics, Linear Algebra, Probability and Statistics, Real Analysis, Fluid Dynamics, General Relativity.

Astrophysics Research Experience

Research Assistant

10/2025 – Present

Raman Research Institute, Bengaluru, India · *Supervisor: Dr. Kartik Sarkar*

- Developed a two dimensional axisymmetric ray tracing radiative transfer simulation that uses discrete ordinate methods for solving the partial differential equation.
- New geometric formulation used in the discrete ordinate method that performs better than conventional methods like S_N quadrature or spherical moments.
- Simulation is written entirely in C++ and is parallelised to provide very quick runtimes with minimal computational resources.

Master's Thesis: Accretion onto Neutron Star Surfaces

08/2024 – 04/2025

Aryabhata Institute of Observational Sciences (ARIES), Nainital, India · *Supervisor: Dr. Indranil Chattopadhyay*

- Developed a novel semi-analytical extension of an in-house Eulerian hydrodynamics code (Fortran / C++ / Python) to simulate accretion onto compact objects with hard surfaces (neutron stars, white dwarfs), adapting existing black-hole accretion solutions to hard-surface boundary conditions.
- Characterised the shock dynamics induced by gas-surface interaction across the parameter space. Work in progress toward journal submission.
- Thesis is [\[linked here\]](#)

Research Internship: Origin of Zodiacal Light

05/2022 – 07/2022

Indian Institute of Science Education and Research (IISER), Mohali, India · *Supervisor: Dr. Jasjeet Singh Bagla*

- Built an N-body simulation in Python / C++ integrating gravitational dynamics and radiation-pressure effects, testing Mars as a candidate source for interplanetary dust. Produced qualitative agreement with observational data.

AI Safety Engagement

Technical AI Safety Project: Can Models Hide Without Being Taught To?

07/2026 – 08/2026

BlueDot Impact · *Part-time cohort, 6 weeks*

github.com/Aswin-R-Nair/activation-hiding

- Investigating whether models can learn to hide their activations from linear probes entirely in-context.
- Experiments were conducted on two scales of models: Gemma-2-9b-it and Llama3.3-70b-Instruct, where the models were tested on zero-shot and few-shot settings to see if they could hide their activations from linear probes on fixed tokens.
- It can be shown that there is a hard prefix attached to the prompts that can be used to obfuscate activations, but the models fail to find them in-context without any explicit training.
- Full blog write-up [\[linked here\]](#)
- **Stack:** PyTorch, HuggingFace, Transformers.

Technical AI Safety Course

05/2026 – 06/2026

BlueDot Impact · *Part-time cohort, 6 weeks*

Independent Project: Behavioural Steering via Linear Probes

Ongoing

github.com/Aswin-R-Nair/truth-vector

- Extracting steering directions from multiple-choice correctness signals (TruthfulQA, MMLU) via mean-difference probes on Qwen2.5, Qwen3, and Llama 3.1.
- Initial hypothesis (the extracted direction encodes truthfulness as a content property) was undermined by cross-model and cross-dataset ablations; current exploratory framing treats the extracted directions as encoding dataset-specific behavioural dispositions (hedging vs. assertion) rather than a unitary truth axis.
- Methodology, current results, and open methodological questions documented in repo README.
- **Stack:** PyTorch, HuggingFace, Transformers, TransformerLens.

Conferences & Workshops

43rd Meeting of the Astronomical Society of India

02/2025

- Poster presentation on Master's thesis research. [\[poster\]](#)

PLUTO Code Symposium

09/2024

- Exposure to the latest release of the PLUTO hydrodynamics package.

GW Open Data Workshop (LIGO)

05/2022

- Methods for isolating, processing, and inferring gravitational-wave detections from raw strain data.

Technical Skills

- **Programming:** Python, C++, Fortran, Mathematica
- **ML / interpretability:** PyTorch, HuggingFace Transformers, TransformerLens
- **Numerical methods:** Eulerian hydrodynamics, N-body simulation, MPI
- **Tools:** Git, Linux, $\text{\LaTeX}$